ledger is also encrypted. This prevents anyone being able to change the data. However, anyone who wants to view the data in the digital ledger is able to do so.

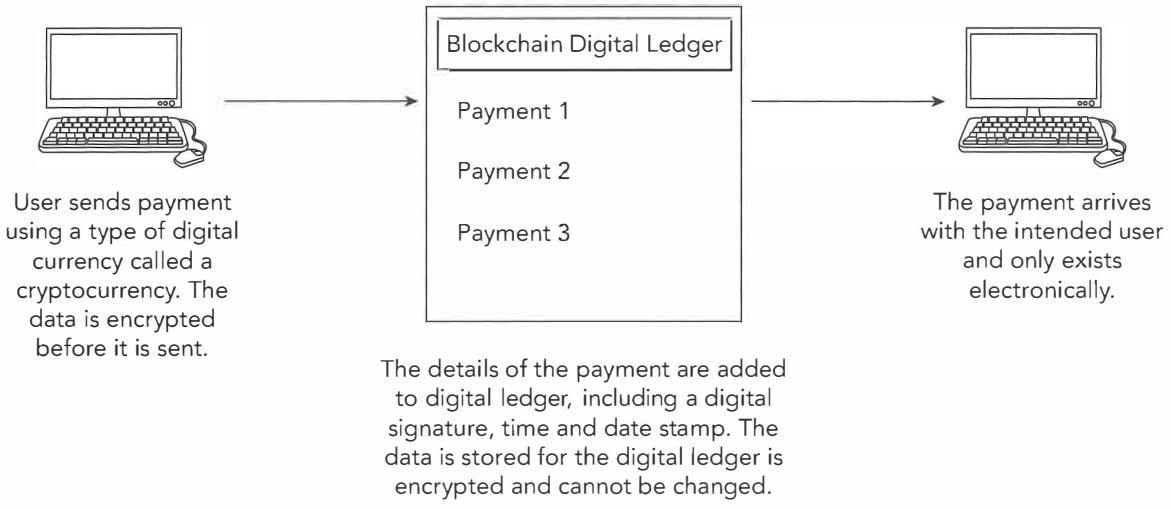


Figure 5.18: A blockchain keeps a digital ledger of transactions

COMPUTER SCIENCE IN CONTEXT

There are as many as 2000 different digital currencies that exist. Many of these are different types of cryptocurrency. How many do you think you could name?

5.5 Cyber security

Whilst the internet and the world wide web can be a very fun and interesting place, it does also have some issues and threats that you definitely need to be aware of. You need to know how these cyber threats are created, what the motivation is behind those who use them, and what kind of security solution you can put in place to try and prevent them from happening. One thing to note is that each security solution may not be a complete prevention, sometimes it is better to have several solutions in place. Also, **perpetrators** are always looking to develop ways to beat security solutions, so no solution will be 100% effective in preventing an attack. However, if you don't have any security solutions in place, you are putting your personal data at a far greater risk of these cyber threats. So, what are the cyber threats that you need to know?

KEY WORDS

perpetrator: a person that carries out an illegal or immoral action.

brute-force attack: a type of cyber threat that involves repeatedly inputting a password until the correct one is found, to try and break into an account or device.

Brute-force attack

You may have heard someone being described as a brute person. This normally means people think they are strong and forceful. This is exactly what a **brute-force attack** is on your computer, a strong and forceful attack. Imagine that someone is banging on a locked door, pushing and shoving it in every way that they can, until the door gives way and opens. In a brute-force attack, a perpetrator tries to break into your computer or account. They do this by repeatedly entering different password combinations until they manage to enter the correct one. They may try to do this manually, by typing in each password combination themselves. They could also do this by developing a

software program that will automatically cycle through many different combinations until the correct one is found.

The perpetrator usually does this to gain access to your computer or account in the hope that they can either steal some personal data that they can use for criminal activity, or they may try to use your account to purchase items online as though it was their own account.

There are several cyber security solutions that you can put in place to help prevent a brute-force attack on your computer or account. The first is to make sure that you have a strong password to protect your data. A strong password is created by mixing different uppercase and lowercase letters, numbers and symbols. Ideally mixing these in a way that does not form any kind of word or phrase. A long string of a mixture of all these characters is the best protection that you can give your data. This will make it much more difficult for any perpetrator or software program to guess your password. It can be easy to find out some simple information about you, for example, that you have a rabbit called Mr Snuffles. Therefore, if your password is mrsnuffles01, this is not going to be difficult for a perpetrator to guess. Make life a bit more difficult for them, your data is precious! An example of a strong password is:

1gH6dk@dnlwm!dosTsjlpa

Many web browsers now have a function that will suggest a strong password for you, mixing a combination of characters like this. They will also save the password for you, so you don't need to keep remembering the string of characters.

Another security solution that can be used to help prevent a brute-force attack is limiting the number of times a password can be entered before the account is locked. You may have had this issue before. You type in what you think you had as a password for the account, but it is rejected, so you try another and that gets rejected, then you try a third one and that is rejected, and you now get a message saying you are locked out of your account. How frustrating! However, be very thankful for this security solution, however annoying it can be. It is likely that for a perpetrator to brute-force their way into your account, it will take more than three guesses at your password. Therefore, if the number of attempts is limited before the account is locked, this can prevent many guesses being made to try and find out the password.

A third security solution that can be used is a **biometric password**. A **biometric device** is one that needs your biological data to be input. An example is a fingerprint scanner, which requires your fingerprint to be input. Our biological data is unique to us, it is highly unlikely that you will find another person that has exactly the same fingerprint as you. Therefore, if you secure your computer or account with a biological password, such as your fingerprint, no amount of guessing your password would allow a perpetrator to brute-force their way into your account. They would need to somehow obtain your fingerprint, which is a much more difficult task.

A fourth security solution that can be used is **two-step verification** (sometimes known as two-factor authentication). This is a method that puts two tasks in place in order to enter an account. The first task is entering the password, but then a second step is added that will require an additional input task to enter the account. The second piece of data is often a code or word that is sent to you that you will need to enter. This is normally sent to either your mobile phone number or your email address. This means that a perpetrator will not only need to guess the password for your account, but that they will also need to have access to your mobile phone or your email, this is an added difficulty. Some banks also have a small card reading device that provide you with the data for the second step. You put your bank card into the reader, enters a PIN code

KEY WORDS

biometric password: a type of password that uses a person's biological data, such as their fingerprint.

biometric device: a device that allows a user to record and input a biometric password.

two-step verification: a process that involves inputting two different kinds of data to enter an account or device.

or password and this generates the code for the second step for you to input. It is advisable that you look at the privacy and security settings for your online accounts, especially ones like your email and social media. Many companies have an option for you to turn on the feature of two-step verification which helps keep your personal data more secure.

In summary:

What happens?	Multiple guesses are made at your password to try and break into your computer or account.
What is the aim?	To steal your precious personal data or to use your account to buy products online.
What security solutions can be used?	A strong password, a limited number of attempts at entering a password, a biometric password and two-step verification.

ACTIVITY 5.4

Think about one of your passwords that you use to access an account online. How strong do you think that password is? Do you think that you could improve that password?

Create a three-slide presentation that can be given to your peers to tell them how to create a strong password and what the importance is of doing so.

Peer Assessment

Take a look at your partner's presentation. Do you think the information is presented clearly? Have they made it clear how to create a strong password? Do you think they have missed any information about doing this? Do you think they have made it clear how important it is to set a strong password? Have they included information about what might happen if a person does not set a strong password?

Tell your partner your answer to these questions when you have looked at their presentation.

Data interception

Billions of pieces of data are sent across the large network that is the internet on a daily basis. A lot of this data is people sending their personal data to do things like buy products and services, and log into their online accounts. Perpetrators would very much like to get hold of their information so that they can use it in criminal activity, such as identity theft and fraud. A way that they will try and do this is by intercepting **data packets** as they are transmitted across the network from one device to another. This is often done by using a piece of software called a **packet sniffer**, this is normally installed onto a piece of hardware that the packets will pass through, for example, a router. This software is used to examine packets of data that are sent across a network to see if they contain data that looks useful. If they do, the software will report the useful data packets it has found back to the perpetrator.

KEY WORDS

data packet: a unit of data that is used to transmit data across a network.

packet sniffer: a piece of software that is used to examine the contents in a packet of data.

The main security solution that you can use to keep data secure that is sent over a network is encryption. Data is normally encrypted using the SSL protocol that you learnt about earlier in Section 5.2. This security solution will not stop the data packets from being intercepted, but it will mean that if the data packets are intercepted, the data contained in the packets will be meaningless to anyone who tries to read them. Remember, you can check to see if a website encrypts your data before transmission by looking to see if the URL uses HTTPS. It is also often possible, in your web browser settings, to set your web browser to only connect to websites that use encryption to transmit data.

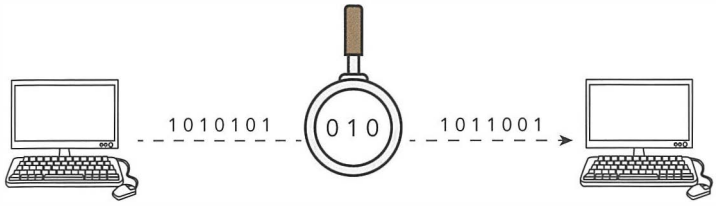


Figure 5.19: If your data is intercepted by someone, will it be encrypted? Check if the URL uses HTTPS

Have you ever gone into a café or a store and thought, 'Excellent, free Wi-Fi!' and you immediately connect to it? Perpetrators can also use this kind of incentive to get you to connect to the network through a system of theirs. Any data that you send using this connection can be intercepted by the perpetrator. It is advisable, if you see a free Wi-Fi connection, to ask the café or store if it is their free Wi-Fi before you connect to it.

In summary:

What happens?	Data packets are intercepted and examined through the use of software such as a packet sniffer.
What is the aim?	To steal your precious personal data for criminal activity such as identity theft or fraud.
What security solutions can be used?	Encryption such as the SSL protocol and visually checking the URL of a website to see if it uses the HTTPS protocol.

ACTIVITY 5.5

Use the internet to find out how a virtual private network (VPN) can be used to help prevent data interception.

Distributed denial of service (DDoS) attack

To carry out a **DDoS attack**, a perpetrator will first create a **botnet** (bot network, see Figure 5.20). This is done by sending **malware** to many computers to try and get you to

KEY WORDS

distributed denial of service (DDoS) attack: a type of cyber threat that targets a web server to cause it to crash and prevent access to the web pages that it stores.

botnet: a network of bots that are created to carry out a DDoS attack.

malware: a type of malicious software that is downloaded onto a user's computer or device.

accidentally download it. Once you have downloaded the malware, the computer can then be used as a **bot** by the perpetrator. Whilst the bot is not being used it is often called a **zombie**. Once the perpetrator wants to carry out the attack, they 'wake up' all the zombies to be used as bots. Each bot will begin sending multiple requests to access a web server. They are all set to send requests to the same web server all at the same time. This floods the web server with too many requests and it begins to struggle to deal with each request. The web server can only deal with a certain number of requests at a time and if it is sent too many it will begin to slow down and eventually crash. This will mean that users can no longer access the website. This causes a denial of service to any normal requests to access the web server. This type of attack is normally carried out on a web server, but it could be used to target other types of server and network hardware.

There are several reasons a perpetrator may carry out a DDoS attack. They may make a demand for money to the company to get them to pay to stop the attack from happening. They may want to punish the company for something that they perpetrator thinks they have done wrong, such as an unethical issue. This would be in an act of revenge or activism.

The main security solution that a company can use to stop a DDoS attack occurring on their web server is to use a **proxy server**. A proxy server can be used as a barrier to a web server by getting all the requests for the web server to pass through the proxy server first. The proxy server can examine each request and forward it on to the web server to allow access. If the proxy server begins to receive many requests, it can begin to allow them to pass at a slower rate to avoid the web server being flooded with all the requests at the same time. This may mean access to the website is slower, but still possible by users that want to legitimately access it. A proxy server can also use a feature called caching. This feature can help prevent a DDoS from being successful as once the proxy server has received a request the access the website from a certain IP address, it will stop any repeated requests for a short period of time from the same IP address. Therefore, if a bot is sending multiple requests, not all of them will be passed onto the web server as some will be stopped by the proxy server. This makes it more difficult for a perpetrator to flood the web server.

As a user, you can try and prevent your computer ever being used as a bot in a DDoS attack by regularly scanning your computer with **anti-malware** software. This is a type of software that scans all of the files in your computer to see if any of them are known to be malware. It does this by comparing each file against a known list of malware to see if any of the files on your computer match. If it finds a file that matches, it will put it in a special area called a quarantine area. It will then inform you that it has found what it thinks is malware and will ask you if you want to delete the file. By scanning your computer with anti-malware software on a regular basis, you can help make sure that your computer is free from any malware that might be used to turn it into a bot in a DDoS attack.

KEY WORDS

bot: the name given to a computer that has had malware downloaded onto it that will cause it to be used in a DDoS attack.

zombie: the name given to a bot that has malware downloaded onto it to allow it to be used in a DDoS attack, but it currently isn't being used in this way. It is effectively a dormant or sleeping bot.

proxy server: a method of cyber security that examines requests sent by devices to access the web pages stored on a web server.

anti-malware: a type of software that scans a computer or device with the purpose of finding and removing malware.

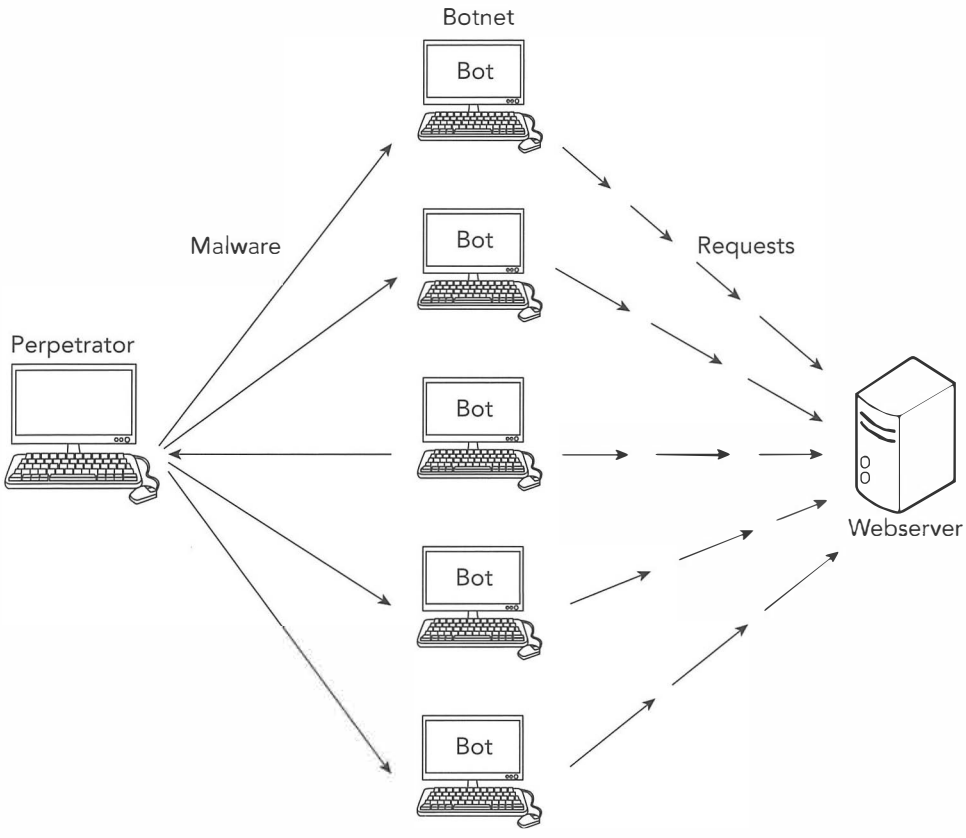


Figure 5.20: A botnet can cause a DDoS attack

In summary:

What happens?	Malware is sent to lots of computers to turn them into bots to create a botnet. The perpetrator then uses the botnet to send many requests to a web server, all at the same time.
What is the aim?	The aim is to cause the web server to crash. The perpetrator may be doing this to demand money for it to stop, or as an act of revenge or activism.
What security solutions can be used?	The company can use a proxy server to act as a barrier that filters requests to the web server. You can scan your computer with anti-malware on a regular basis to try and prevent it being used as a bot.

ACTIVITY 5.6

Use the internet to find out about a huge DDoS attack that was carried out on 21st October 2016, and which companies it affected.

Hacking

Hacking is the act of trying to gain unauthorised access to data. A perpetrator that does this is known as a **hacker**. They can do this by using a range of technical skills that allow them to break into a computer or a network. This can be done by going through methods such as a brute-force attack (this is sometimes known as cracking) or by exploiting vulnerabilities in a computer or a network, which are areas that the hacker can identify that lack the protection of security solutions.

Hackers will gain access to a system for several reasons. This could be to steal your data to try and steal your identity, as an act of revenge, to corrupt your data or to leak confidential data about a company or project, or an act of activism.

The main way that you can stop your computer being hacked is to have as many security solutions in place as possible. One of the main security solutions that can be used is a **firewall**. A firewall is a barrier that can be used between your data and the rest of a network. It can be software based or hardware based. Any data that is sent into and out of your computer is examined by the firewall. Rules can be set for this data by giving your firewall a set of criteria to examine the data against. The data that is sent through the firewall is compared to the criteria. The firewall can be set to accept the data and allow it to pass through the firewall, or reject the data and stop it from passing through the firewall, depending on whether it does or does not meet the criteria. This added security solution may help identify data that a hacker has sent that looks malicious and reject this data to stop the hacker getting into the computer or network. A firewall can also be used to close certain **ports** to a network. A hacker may look to find a vulnerability, such as an open port, to get into a network. A firewall can be set to close as many ports as it can that are not being used, to try and prevent a hacker exploiting an open port to gain access.

One kind of vulnerability that hackers may try to exploit is out of date software. Sometimes, a software company may find that they have an issue with their software that could allow a hacker to gain unauthorised access. If you have downloaded this software, you have also created this vulnerability on your computer. As software companies become aware of these issues, they develop their software and issue updates to protect against hackers. It is therefore vital that you update your software on a regular basis when software updates are released. This helps limit any vulnerabilities you have on your computer that a hacker can exploit. A way to make sure that your software is always up to date is to set your software to automatically update. This means that as soon as a software update is released, your computer will automatically download the update and you don't need to remember to do this.

You can also password protect all your data with strong or biometric passwords to help stop a hacker gaining unauthorised access to your data. And you can add two-step verification to all your accounts.



Figure 5.21: Good security will protect computers against hackers

In summary:

What happens?	A hacker gains unauthorised access to your data by exploiting a vulnerability in your computer or network.
What is the aim?	The aim is to steal, corrupt or leak your data for criminal activity.
What security solutions can be used?	A firewall, automatic software updates, strong or biometric passwords and two-step verification.

KEY WORDS

hacking: the act of gaining unauthorised access to data. This is normally done to steal or corrupt the data.

hacker: the name given to a perpetrator that carries out an act of hacking.

firewall: a cyber security method that is used to examine incoming and outgoing traffic from a computer or network.

port: an entry point into a computer or network.

Questions

9 What is a hacker?

10 How can you prevent a hacker gaining access to your data?

Malware

Malware is a term used to describe any malicious software that is designed to disrupt your computer or data. There are several types of malicious software that you need to know about: **virus**, **worm**, **trojan horse**, **spyware**, **adware** and **ransomware**. See Table 5.1.

Malware	Description
Virus	This is a computer program that is downloaded onto your hard drive. Once it has downloaded it replicates itself and corrupts your stored data or uses up all the available memory in your computer, causing it to slow down and crash.
Worm	This is a computer program that also replicates itself. It has a different aim to a virus though. It looks to find vulnerability holes in a network to use to replicate itself. In doing this it will clog up the bandwidth of a network and slow the network down.
Spyware	This is a computer program that is downloaded onto your hard drive and is designed to record your actions on your computer. A common form of spyware is a keylogger. This records all the key presses that a you carry out on your keyboard. This data is then sent to the perpetrator where it can be analysed to identify patterns in your data. Patterns in your data could reveal personal data such as your passwords. This can allow access to your online accounts for fraud and identity theft.
Trojan horse	This is a computer program that is used to disguise other malware. It is designed to look like a harmless piece of software, such as an application or game, but it contains other malware such as a virus or spyware. Once the trojan horse is downloaded onto your hard drive, it releases the other malware that it contains.
Adware	This is a computer program that is designed to automatically created pop up and banner adverts when you are online. These adverts are often unwanted and can be irritating and frustrating. The adware creators are given money by the companies that appear in the advertisements when you click on them.
Ransomware	This is a computer program that is downloaded onto your hard drive or other hardware. It is designed to encrypt your data and stop you from gaining access to it. The creators of the ransomware will demand a ransom (a set amount of money) for the data to be decrypted. They will often threaten to release and leak all your stored data if the ransom is not paid.

Table 5.1: Different types of malware

KEY WORDS

virus: a software based cyber threat that replicates itself with the aim of corrupting data or filling up the available memory space in a computer, causing it to crash.

worm: a software based cyber threat that replicates itself across a network using vulnerabilities that it finds, with the aim of clogging up the bandwidth.

trojan horse: a software based cyber threat that is used to disguise other malware to try and smuggle it into a computer or network.

spyware: a software based cyber threat that spies on a user's action whilst using a computer, such as logging their key presses.

adware: a software based cyber issue that automatically creates popup advertisements.

ransomware: a software based cyber threat that encrypts a user's data to stop them gaining access to it until a ransom (money) is paid.

There are several security solutions that you can use to protect against the threat of the different types of malware.

An **anti-virus** software can be used to scan your computer for viruses. This is a type of software that scans all of the files in your computer to see if any of them are known to be a virus. It does this by comparing each file against a known list of viruses to see if any of the files on your computer match. If it finds a file that matches, it will put it in a special area called a quarantine area. It will then inform you that it has found what it thinks is a virus and will ask you if you want to delete the file. By scanning your computer with anti-virus software on a regular basis, you can help make sure that your computer is free from any viruses. It is also important to update your anti-virus software. Perpetrators are constantly developing new viruses. The anti-virus software developers are constantly finding these and update their database used in the scan. You need to make sure that you are updating your anti-virus to make sure that the database is up to date and all the viruses are found.

KEY WORD

anti-virus: a type of software that scans a computer for viruses with the purpose of finding and removing them.



Figure 5.22: Anti-virus software should be kept up to date to find any and all viruses

Anti-malware software can be installed and used to scan your computer and network for other types of malware, such as worms, spyware, adware and ransomware. This operates in a similar way to the anti-virus software by scanning, quarantining and deleting any malware found. The anti-malware that looks specifically for spyware is called anti-spyware.

A firewall can be used to help prevent malware being downloaded onto your computer. The data coming into and out of your computer can be examined against the criteria set for the firewall. If the data looks malicious, the firewall will reject it and stop it from being downloaded onto your computer. It is useful that a firewall examines both the data incoming and outgoing data from your computer, especially in the prevention of spyware. As the firewall is examining the data leaving your computer as well, it may recognise a malicious program that is trying to leave your computer that contains all the keypresses that you have made on your keyboard. If the firewall rejects this data, it will not be able to leave your computer, preventing it being sent to the perpetrator to be analysed.

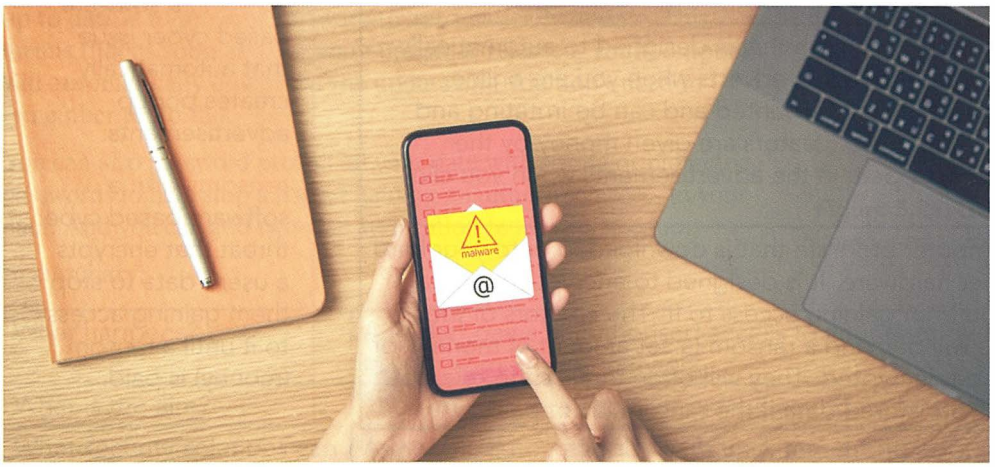


Figure 5.23: Malware can be sent to a computer via email

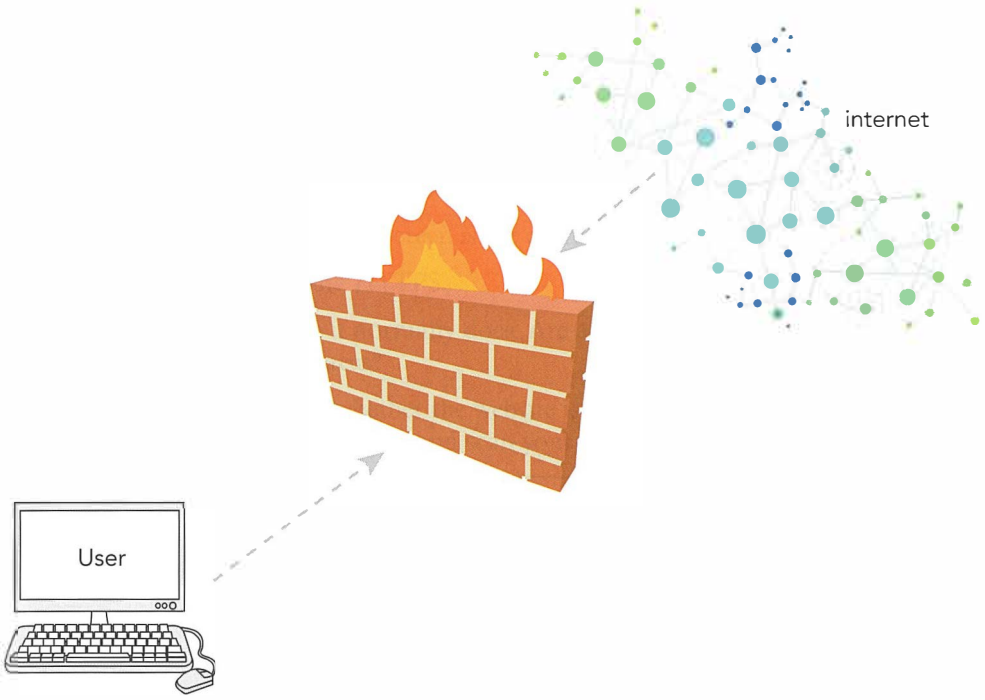


Figure 5.24: A firewall protects a user's computer from security threats on internet

You can make sure that you are very cautious when clicking any links and downloading any software online. Malware is often hidden in what looks like an innocent piece of software or link, for example, a game that looks enticing to play. There could be malware hidden inside a trojan horse in the game that may not be detected by a firewall. If you download the game, it will download the trojan horse, allowing other malware to be released onto your computer. You should only ever download software from companies and sources that you know can be trusted, no matter how fun or enticing it may look.

You can also store a **backup** of your data that is not attached to your computer or network on a constant basis. This means that if your data is damaged by malware, or encrypted using ransomware, you still have access to a copy of your data. This can prevent you needing to pay a ransom to gain access to your data again, but it will not stop the perpetrator from leaking your data. Therefore, it is advised not to keep any personal data on your computer that could be accessed and leaked by a perpetrator of ransomware. If you do keep personal data stored on your computer, you should encrypt the data yourself, so if the perpetrator did gain access to the data, it would be meaningless to them and they wouldn't be able to leak it.

To avoid the risk of personal data, such as passwords, being sent to a perpetrator, you could use an onscreen virtual keyboard rather than a physical one to enter passwords. This means that you don't need to make any key presses, so these are not recorded for your password, so any data that was sent to a perpetrator would not contain your passwords. Some companies also add drop down boxes for you to select characters from to input your password. They may also ask you for random characters from your password, and not your whole password. This is to prevent you needing to enter your full password and to stop your full password being sent to the perpetrator.

KEY WORD

data backup: a copy of data that is stored separate from the computer.

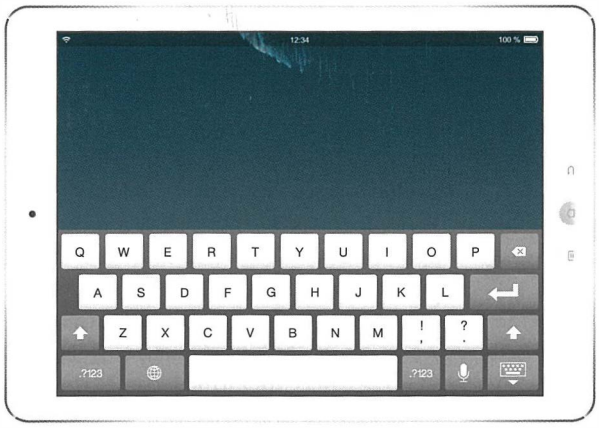


Figure 5.25: An onscreen virtual keyboard

In summary:

What happens?	Malicious software is downloaded onto your computer or network.
What is the aim?	Malicious software is used to corrupt your data, gain access to your data or damage your hardware.
What security solutions can be used?	Anti-malware software, including anti-virus and anti-spyware, a firewall, exercising caution when downloading software and creating a backup of data.

Pharming

Pharming is used by perpetrators to try and get your personal data, such as your username and password for your online accounts. They can then log into the account, pretending to be you, and use your account to buy products online with your money. It can also be used to get more personal details from you, such as your address, that can be used by the perpetrator for identity theft and fraud.

The perpetrator will try and entice you into clicking a link or icon to start a download. This could be done in various ways including sending an email, creating a pop up advert or just having a link available on a web page. When you click the link, a download will begin and as part of that download, malicious software will be downloaded onto your **hard drive**. This malicious software is designed to redirect certain website requests you have to go to a fake website instead (see Figure 5.26). The perpetrator will set up a fake website to look very similar to a genuine website, such as PayPal or your social media page. When you next type in the URL to request the web page, the malicious code that has been downloaded on your hard drive will redirect the request to the fake website instead. The perpetrator will hope that you think you have opened the genuine website and when you type in your login details, these will be sent to the perpetrator instead.

If you notice that the website is fake after typing in your login detail, it is very important that you change your password for that website immediately. This can help stop the perpetrator being able to log into your account.

As pharming uses malicious code that is installed on your hard drive, you can use anti-malware software to remove it. The anti-malware software will scan the files stored on

KEY WORDS

pharming: a type of cyber threat that involves downloading malicious software onto a user's hard drive, that redirects a request aimed at a genuine website to a fake website instead. The fake website will encourage the user to input their personal data, allowing it to be stolen.

hard drive: a storage device that stores data permanently, until it is deleted by the user.

your computer and will help to find the file that is redirecting you to the fake website. The anti-malware software will quarantine this file and let you delete it. When you type in the URL now the redirection should stop and you should be taken to the genuine website.

One thing that you can do to help prevent pharming is to only download software or click links that you know are from a trusted source. You should also thoroughly check any website before you enter your personal details. You can look to see:

- If the URL at the top of the screen is the correct one.
- If the site is secured using HTTPS.
- If the website looks like it normally does. Are all the usual colours, images and text there in the same places?

If you are in any doubt at all whether the website is genuine, close the website without entering any of your details and scan your computer immediately with anti-malware software.

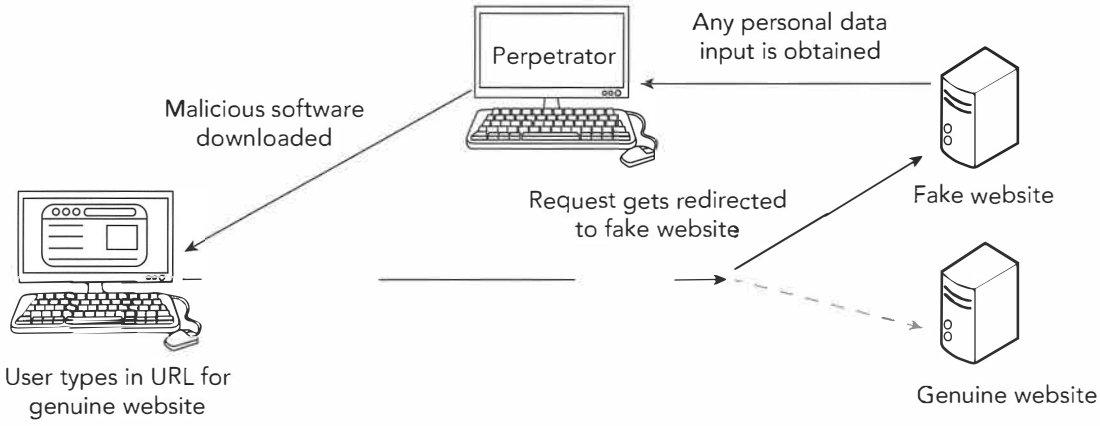


Figure 5.26: Pharming is a cyber threat where fake websites are used to steal data

In summary:

What happens?	Malicious software is downloaded onto your hard drive that will redirect you to a fake web page when you input the genuine URL.
What is the aim?	To get your personal data to commit criminal activity such as stealing your money, identity theft or fraud.
What security solutions can be used?	Anti-malware software, visually checking the web page and only downloading data from trusted sources.

Question

11 What can you do to prevent a perpetrator getting your personal data through pharming?

Phishing

Phishing is also used by perpetrators to try and get your personal data, such as your username and password for your online accounts. They can then log into the account,

KEY WORD

phishing: a type of cyber threat that involves sending a user a fake email that is designed to look genuine. It will encourage the user to provide their personal data either by clicking a link to a fake website, or by responding to the email.

pretending to be you and use your account to buy products online with your money. It can also be used to get more personal details from you, such as your address, that can be used by the perpetrator for identity theft and fraud.

The perpetrator will again try and entice you into clicking a link. In phishing, this is done by sending you an email. The perpetrator will create an email that looks like it is from a genuine company, such as your bank. The email will try and get you to click a link that will take you to a website. The email will try and get you to click the link by telling you things like:

- It looks like there has been unusual activity on your account, please click the link and log in to check.
- You have won a prize, please click the link and enter your details to claim the prize.
- Confirmation is needed that you have made this purchase, please click the link and log into your account to confirm whether or not you have made this purchase.
- You are being paid a refund on your taxes, please click the link and enter your credit card details so that you can claim the refund.

The link will take you to website that will not be the genuine website, it will be a fake website that has been made to look like the genuine one. If you then input your personal details these will be sent to the perpetrator and they will be able to access your account or try and build a profile to steal your identity.

You can help prevent yourself getting fooled by any phishing scams by checking:

- The spelling and tone of the text in the email. Is everything spelt correctly? Is the email written in a professional tone? Most genuine companies will spend a lot of time making sure their emails are professionally written.
- Whether the email has been personalised or if it is addressed in a generic way, e.g. Dear customer. Many phishing emails are sent out in bulk and this often means they are not individually addressed.
- The URL that is attached to the link. If you hover your mouse pointer over the link, you should be able to see the URL that the link is connected to. Check if this is the correct URL for this company.
- If the URL at the top of the screen is the correct one.
- If the site is secured using HTTPS.
- If the website looks like it normally does. Are all the usual colours, images and text there in the same places?

Most companies will not ask you to provide your personal details over the internet in this way. If you are in any doubt at all whether the website is genuine, close the website without entering any of your details. If you have entered any of your details and realise after that you think it is a phishing scam, make sure that you change the password to your account and any other accounts that use that password immediately.



Figure 5.27: In a phishing scam perpetrators will try to get your password

Question

12 What are three things that you can look for to identify if an email is a phishing scam?

Social engineering

Social engineering relies on manipulating and deceiving people into providing confidential or personal data that can be used in criminal activity such as identity theft and fraud, and hacking into computers systems or networks. Social engineering can take many different forms. Phishing is also one form of social engineering, in that it manipulates you into providing your personal data.

KEY WORD

social engineering: a cyber threat that involves manipulating or deceiving people into providing confidential or personal data.

Have you ever seen one of those quizzes on social media that ask you to provide lots of fun details about you, like your favourite colour, the name of your pet and your favourite food? These quizzes are often created by perpetrators trying to find out key information about you. Many people base their passwords around this type of information, so a perpetrator can gather lot of information like this about you every time you fill in one of these quizzes on your social media. They can then use this data along with a brute-force password program to try and get into your online accounts. Advice for the future, do not fill out these quizzes! This is another form of social engineering, it manipulates and deceives you into thinking you are filling out a fun quiz to learn more about your friends, but you are actually providing key information that a perpetrator can use against you. One way that you can limit a perpetrator's access to your data on social media is to have your privacy settings set to the highest level of privacy. This will normally mean that only people that you are connected to will be allowed to see your data.

Social engineering can be used to target companies for their data too. A perpetrator could make contact with the company and try and manipulate and deceive employees into providing key information that can be used to hack the company systems. Here are two examples of how this could be done:

Example 1

Large companies often have quite a large IT department that are often not too well known by all the employees. It is a common occurrence that an employee may have an IT issue with their computer in a morning. They will normally report this issue to the IT department in their company and will be told someone will contact them from the department, shortly. Therefore, a perpetrator can try and use this situation. They get hold of a list of all the telephone numbers of the employees in the company and begin to call them. They will say that they are from the IT department in the company and say that they are aware of the IT issue that they have reported. Most employees will probably just say they haven't reported an issue, but if they happen to call an employee that has, that employee will think it is the IT department calling to sort their problem. The perpetrator will then say that they just need the employees login details to access their system to see what is wrong with it. The employee may be tricked into providing their login details and

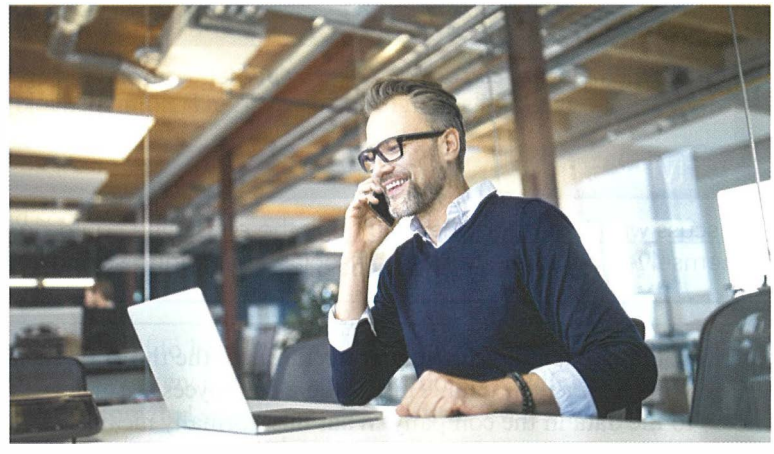


Figure 5.28: Employees of large companies can be contacted by perpetrators claiming to be from the IT department

now the perpetrator can get into the computer systems of the business and steal any data that they think could be useful to them, or look to introduce malware.

ACTIVITY 5.7

Discuss with a partner what you think the employee could do to stop themselves being the target of a social engineering scam like this one.

Example 2

Employees often go for dinner together or out for a relaxing coffee after work on a Friday. A perpetrator finds out where employees do this for the company that they want to target. The perpetrator places themselves close to where the employees will normally sit. They will start a conversation with the employees and become friendly with them. They will start to ask lots of questions about the company an employee works for. The employee may mistake this as the person taking an interest in their life. Over a few what seem like accidental meetings with the employee the perpetrator has asked lots of questions that has allowed them to build up a profile about the company, such as information about where they store their data and what kind of security systems they have. The perpetrator can use this information to hack into the company systems and network.



Figure 5.29: Employees can be contacted by perpetrators pretending to be colleagues

ACTIVITY 5.8

Discuss with a partner what you think the employee could do to stop themselves being the target of a social engineering scam like this one.

One way that companies can help protect against the threat of social engineering is to use **access level** for their data. This is where employees are given different levels of access to the data in the company. What this normally means is that an employee will only have access to the data that they directly use on a daily basis. It also often means that the only people that have access to more confidential data are the more senior employees in the company. Levels of access are linked to an employee username.

KEY WORD

access level: the amount of direct access a user is given to data and resources. This could be set to be only certain sections of a whole collection of data and resources.

REFLECTION

How did you come to a decision about what the employees could have done in Activities 5.7 and 5.8?

The **username** will have data connected to it that will show the computer system which data the employee is allowed access to. This means that if an employee is targeted by a social engineering scam and fooled by it, the data that the perpetrator can gain access to may be limited. A company can also train employees to recognise social engineering to make them more aware of when it might be happening to them.

In summary:

What happens?	A perpetrator will try and manipulate and deceive you into providing them with personal or confidential data.
What is the aim?	To get data to commit criminal activity, such as stealing your money, hacking into a computer network and identity theft or fraud.
What security solutions can be used?	Access level and awareness of how social engineering is conducted.

KEY WORD

username: a text-based identification that is given to a user that identifies the level of access that they have to stored data and resources.

SUMMARY

The internet is the infrastructure that is used to access the world wide web.
The world wide web is the collection of all the web pages that are available.
A URL is a text based address for a web page that contains the protocol, the domain name and the web page name or file name.
The HTTP protocol is used to transmit data between a computer and a web server. This data is not encrypted when sent using this protocol.
The HTTPS protocol is the encrypted version of the HTTP protocol. It uses the SSL protocol to encrypt the data.
The main purpose of a web browser is to render HTML to allow a user to view web pages. It also has other functions such as recording a user's history and storing bookmarks, favourites and cookies.
Cookies are used to save your personal data and track your online preferences. They can be session or persistent based.
Web pages are located, retrieved and displayed using a web browser, a DNS and a web server.
A digital currency is a currency that only exists electronically. An example can be cryptocurrency and this kind of digital currency is tracked using blockchaining. This is a type of digital ledger that records all payments made using the currency.
There are several cyber security threats that exist such as brute-force attacks, data interception, DDoS, hacking, malware, pharming, phishing and social engineering.
The aim of most of the cyber security threats is to steal your personal data to commit identity theft and fraud, or as an act of revenge or activism.
There are a range of cyber security solutions that can be used to help keep your data safe from security threats such as anti-malware software, firewalls, proxy servers, encryption, authentication (including biometric passwords and two-step verification), privacy settings, automated software updates, access levels and visual checks such as checking the URL that is connected to a link or download.